

DEPARTMENT OF ECE — DIGITAL DESIGN & COMPUTER ARCHITECTURE

Module 2: Combinational Logic & ALU Architecture

From Boolean Minimization to 32-Bit RISC-V Arithmetic Logic Units

Course Code: 26EC1202E

Combinational Logic Foundations

Core principles governing memoryless compute blocks in digital systems



Zero Feedback Loop

Outputs depend strictly on present inputs. No storage elements, no memory states, and zero clock synchronization requirement.



Basic Gate Assemblies

Constructed directly from standard AND, OR, NOT, XOR, and NAND gates forming simple mathematical logic paths.



RISC-V Building Blocks

Forms the fundamental compute engine inside processors including ALUs, MUX selectors, Decoders, and Comparators.

Expression Minimization with K-Maps

Visual reduction technique optimizing logic gate counts for high-speed hardware

Gray Code Order

Cells ordered as 00, 01, 11, 10 to guarantee adjacent cells differ by exactly 1 bit.

Power-of-2 Grouping

Group adjacent 1's in rectangular sizes of 1, 2, 4, 8, or 16 cells to simplify SOP.

Wrap-Around Rules

Edges and corners connect continuously to form maximum possible prime implicants.

Example: 4-Variable K-Map SOP

Minimal Sum-of-Products derivation eliminates redundant gate logic across multi-bit binary adders.

$$F(A,B,C,D) = \Sigma(0, 2, 8, 10)$$

Minimized Output: $F = B' \cdot D'$

Binary Arithmetic Core: Adders

Fundamental circuits executing binary addition in digital processor ALUs

Half Adder (HA)

Adds 2 single-bit inputs without prior carry in.

Sum Output: $S = A \oplus B$ (XOR Gate)

Carry Output: $C_{out} = A \cdot B$ (AND Gate)

Limitation: Cannot accept carry input from lower bits.

Full Adder (FA)

Adds 3 bits: Input A, Input B, and Carry-In (C_{in}).

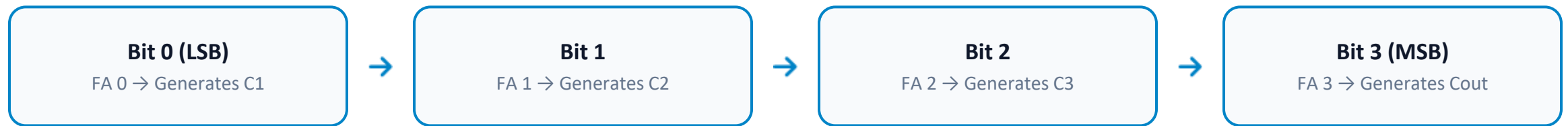
Sum Output: $S = A \oplus B \oplus C_{in}$

Carry Output: $C_{out} = (A \cdot B) + C_{in} \cdot (A \oplus B)$

Built using 2 Half Adders + 1 OR Gate.

Ripple Carry Adder (RCA) Delay Analysis

Cascading full adders for N-bit operands and propagation delay constraints



Total Propagation Delay Formula

$t_{\text{delay}} = N \times t_{\text{FA}}$ (where N is bit-width and t_{FA} is single FA delay)

Linear $O(N)$ Propagation

Controlled Binary Adder / Subtractor

Utilizing Two's Complement arithmetic via XOR conditional inversion

Addition Mode (SUB = 0)

XOR gates pass operand B unchanged ($B \oplus 0 = B$). Initial carry $C_{in} = 0$. Circuit computes $A + B$.

Subtraction Mode (SUB = 1)

XOR gates invert operand B ($B \oplus 1 = B'$). Initial $C_{in} = 1$ adds +1, completing 2's complement ($A + B' + 1$).



Hardware Reuse

Zero extra adders required. Single control wire toggles arithmetic operations instantly in RISC-V ALUs.

Data Routing: Multiplexers (MUX)

Hardware data selectors routing multiple data channels into single outputs

2-to-1 MUX

1 Select Line (S0)

Routes Register vs Immediate values into RISC-V ALU input B.

4-to-1 MUX

2 Select Lines (S1, S0)

Selects final output among ADD, SUB, AND, and OR computation results.

8-to-1 MUX

3 Select Lines (S2, S1, S0)

Routes complex opcode operations across full 32-bit RISC-V instruction set.

Operation Decoding: Decoders

Converting binary code inputs into active-high one-hot signal lines

2-to-4 Binary Decoder

Translates 2 input lines into 4 unique output lines. Exactly 1 output line is HIGH for any input combination.

3-to-8 Control Unit Decoder

Decodes 3-bit RISC-V function fields (funct3) to enable specific hardware functional execution units.

Truth Table: 2-to-4 Decoder

A1	A0	Y3	Y2	Y1	Y0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

Digital Magnitude Comparators

Evaluating quantitative relationships between binary operands: $A > B$, $A = B$, and $A < B$

Equality ($A = B$)

$$(A_0 \odot B_0) \cdot (A_1 \odot B_1)$$

Uses XNOR gates to evaluate matching bit pairs. High only when both operands are identical.

Greater Than ($A > B$)

$$A_1 \cdot B_1' + (A_1 \odot B_1) \cdot A_0 \cdot B_0'$$

Checks MSB first. If MSBs equal, evaluates LSBs for magnitude supremacy.

Less Than ($A < B$)

$$A_1' \cdot B_1 + (A_1 \odot B_1) \cdot A_0' \cdot B_0$$

Drives RISC-V SLT (Set-Less-Than) conditional branch operations in hardware.

1-Bit Multi-Operation ALU Cell

Unifying logic gates, full adder, and MUX selection into a single bit-slice

Operation Map (2-Bit Control)

Op = 00

Bitwise AND Operation

Op = 01

Bitwise OR Operation

Op = 10

Addition / Subtraction (Adder Block)

Op = 11

Set Less Than (SLT Comparison)

Bit-Slice Modular Design

Repeat this 1-bit logic cell 32 times and cascade carry signals (Cout → Cin) to build a complete 32-bit RISC-V CPU ALU.

Complete RISC-V Datapath Integration

Integrating the ALU into register files, control unit, and memory buses

1. Operand Fetch

Registers x0–x31 supply operands A and B directly to the ALU inputs in a single clock cycle.

2. Parallel Compute

ADD, SUB, AND, OR, and SLT sub-circuits evaluate inputs concurrently in hardware.

3. Result Selection

ALU Control decoder drives MUX selector lines, routing desired result to Write-Back bus.

Module 2 Key Takeaways

Combinational logic forms the core execution layer of digital CPUs, translating raw logic gate equations into full arithmetic processors.

K-Map Minimization

Controlled Subtraction

RISC-V ALU Datapath